Automatic Category Suggestion for Wikimedia Commons — Final Report

Megathon · Oct 11, 2026

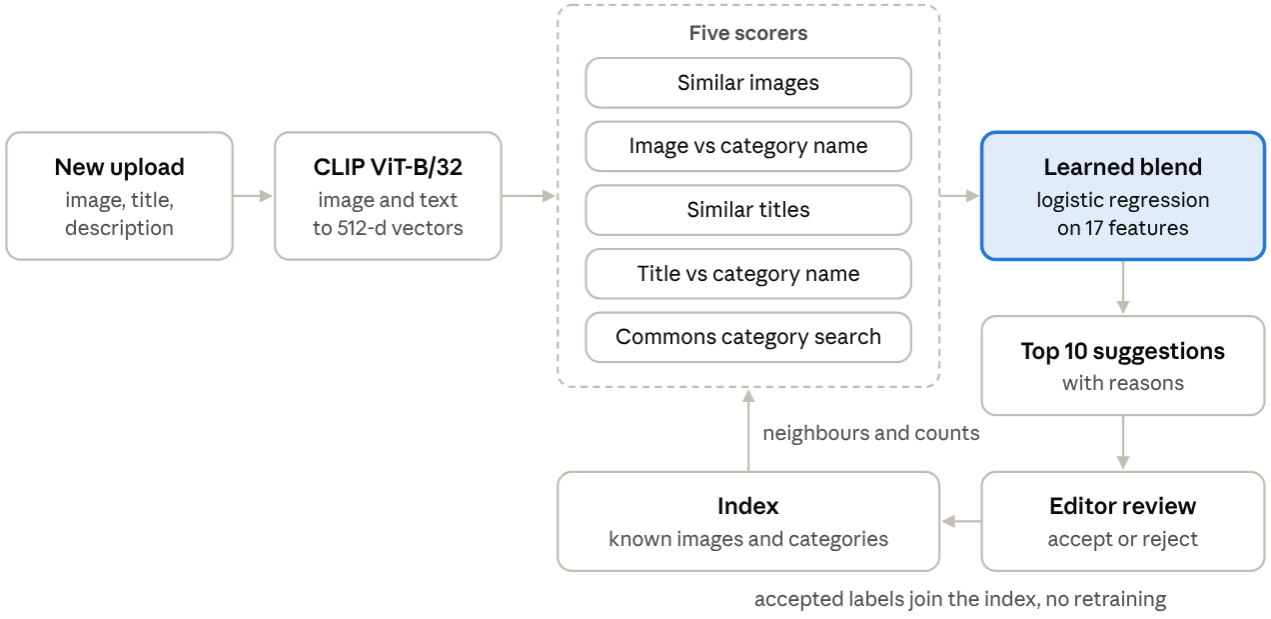
Solution

We built a system that suggests Wikimedia Commons categories for a new upload. On 2,363 held-out recent uploads, its top suggestion is a category editors actually used **50.7% of the time**, and **53.8%** of the true categories appear in its top 10.

Approach

Five scorers each rank candidate categories, and a learned blend combines them. All settings are tuned on validation data; test is used only for the final numbers.

Five signals, one learned blend, and feedback that updates the index at once



system pipeline · 5 scorers, 1 learned blend, feedback loop

Data. 11,962 images with their non-hidden categories, titles and descriptions, fetched through the MediaWiki API. They come from 43 seed categories (object, scene, event, abstract) plus random uploads. Workflow and attribution categories such as "Photos by …" are dropped. OpenCLIP ViT-B/32 (LAION-2B) embeds each image, title and category name.

Split. We split by upload time, as in real use: 8,234 training, 1,165 validation and 2,363 test images. Test is every upload from 24 March 2025 on. Near-duplicates (CLIP cosine similarity above 0.95) are removed across splits.

Five scorers.

1. **Similar images:** the 20 nearest training images vote with their categories.
2. **Image vs category name:** CLIP zero-shot match of the photo to each name.
3. **Similar titles:** the same vote, using title and description vectors.
4. **Title vs category name:** the text matched to each name.
5. **Commons search:** live category search on title words, which finds categories never seen in training.

Learned blend. A logistic regression, fit on validation, re-ranks each scorer's top 30 candidates. It uses 17 features: per scorer the score, log-score and inverse rank, plus category frequency and a search-only flag. The neighbour count and temperatures were tuned on held-out validation.

Features

- **Query** by a random test image, a Commons file name or link, or an uploaded photo.
- **Open vocabulary:** new categories found by search are marked as such.
- **Explanations:** each suggestion shows its image, title and search shares and the similar images behind it.
- **Instant feedback:** accepted labels join the index at once, and decisions are logged for the next fit.
- **Light to run:** one 151 M-parameter model and a 34 MB index, on CPU, CUDA or Apple Silicon.

Results

Combining all five signals doubles top-1 accuracy over the image alone, from 0.242 to 0.507, on the 2,363 test images. The learned blend adds 4 points over fixed weights. Removing workflow categories and tuning added 1 more, from 0.496.

Method	P@1	P@5	R@5	R@10	F1@1	mAP	R@10, non-seed labels
Similar images	0.223	0.064	0.187	0.209	0.154	0.800	0.020
Image vs category name	0.052	0.033	0.102	0.160	0.038	0.828	0.104
Similar titles	0.318	0.084	0.250	0.258	0.229	0.811	0.012
Title vs category name	0.251	0.092	0.335	0.389	0.204	0.854	0.335

Method	P@1	P@5	R@5	R@10	F1@1	mAP	R@10, non-seed labels
Commons search	0.260	0.075	0.247	0.268	0.201	0.795	0.355
Image only (2 image scorers)	0.242	0.076	0.225	0.261	0.170	0.829	0.077
All five, fixed weights	0.465	0.150	0.482	0.526	0.358	0.865	0.359
All five, learned blend (final)	**0.507**	**0.155**	**0.498**	**0.538**	**0.390**	**0.866**	**0.371**

P@k is the share of the top k that editors used. R@k is the share of true categories found in the top k. Non-seed R@10 counts only categories outside the 43 seeds. P@5 is low because most images have only 1 to 3 true categories.

By kind of image. Scenes are easiest; random uploads are hardest. The text and search signals add the most where the image alone is weakest.

Image kind	Test images	P@1	R@10	F1@1	Image-only P@1
Scene	301	0.850	0.698	0.657	0.621
Event	410	0.698	0.567	0.466	0.417
Abstract	131	0.641	0.449	0.387	0.466
Object	403	0.541	0.628	0.390	0.261
All seed images	1,245	0.678	0.606	0.479	—
Random uploads	1,118	0.317	0.462	0.291	0.044

Robustness. We degraded each test image, then re-embedded and scored it with the final model. P@1 fell by at most 1.2 points from 0.507 clean: 0.495 with blur, 0.500 at JPEG quality 10, and 0.502 / 0.499 at 96 / 48 px. Text and search carry the ranking when the image is poor.

Speed. One upload takes about 0.86 s end to end (median, Apple Silicon laptop), and 813 ms of that is the Commons search call. CLIP takes 23 ms for the image and 14 ms for the text, and scoring takes 9 ms. In batches the GPU handles 110 images/s, or 9.5 M a day.

Limitations and next steps

- **Search is slow and partial.** It takes 95% of latency and finds only 35% of new categories. Caching and better queries are the biggest next gain.
- **Not yet used:** the category hierarchy, and a larger or multilingual CLIP. At 100 M images, an approximate-nearest-neighbour index (about 13 GB) would replace brute-force search.